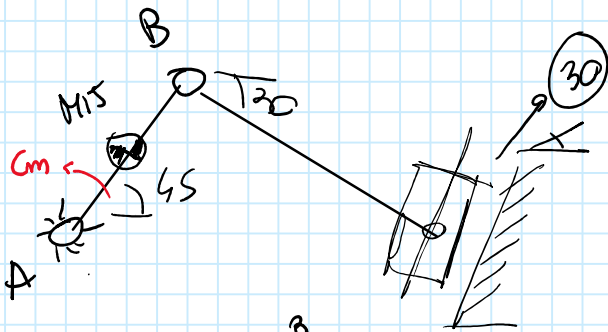
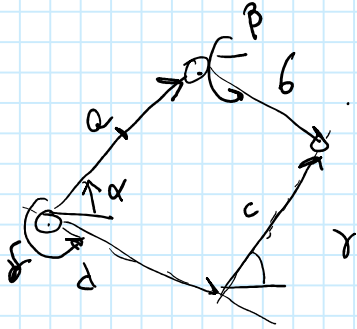




1) $\vec{V}, \vec{a}$ della massa M_c

2) C_m pu avere alto di
Hato



$$a e^{i\varphi} + b e^{i\beta} = c e^{i\gamma} + d e^{i\delta}$$

$$\begin{cases} a \cos \varphi + b \cos \beta = c \cos \gamma + d \cos \delta \\ a \sin \varphi + b \sin \beta = c \sin \gamma + d \sin \delta \end{cases}$$

$$\text{vel: } a \dot{\varphi} e^{i\varphi + \pi/2} + b \dot{\beta} e^{i\beta + \pi/2} = \dot{c} e^{i\gamma}$$

$$\begin{cases} -a \dot{\varphi} \sin \varphi - b \dot{\beta} \sin \beta = \dot{c} \cos \gamma \\ a \dot{\varphi} \cos \varphi + b \dot{\beta} \cos \beta = \dot{c} \sin \gamma \end{cases}$$

$$\begin{cases} b \dot{\beta} \sin \beta + \dot{c} \cos \gamma = -a \dot{\varphi} \sin \varphi \\ b \dot{\beta} \cos \beta - \dot{c} \sin \gamma = -a \dot{\varphi} \cos \varphi \end{cases}$$

$$\begin{bmatrix} b \sin \beta & \cos \gamma \\ b \cos \beta & -\sin \gamma \end{bmatrix} \begin{Bmatrix} \dot{\beta} \\ \dot{c} \end{Bmatrix} = \begin{Bmatrix} -a \dot{\varphi} \sin \varphi \\ -a \dot{\varphi} \cos \varphi \end{Bmatrix} \quad \begin{cases} \dot{\beta} = 0.97 \\ \dot{c} = 1.9 \end{cases}$$

$$\text{accel: } a \ddot{\varphi} e^{i\varphi + \pi/2} - a \dot{\varphi}^2 e^{i\varphi} + b \ddot{\beta} e^{i\beta + \pi/2} - b \dot{\beta}^2 e^{i\beta} = \ddot{c} e^{i\gamma}$$

$$\begin{cases} b \ddot{\beta} \sin \beta + \ddot{c} \cos \gamma = -b \dot{\beta}^2 \cos \beta - a \ddot{\varphi} \sin \varphi - a \dot{\varphi}^2 \cos \varphi \\ b \ddot{\beta} \cos \beta - \ddot{c} \sin \gamma = +b \dot{\beta}^2 \sin \beta - a \ddot{\varphi} \cos \varphi + a \dot{\varphi}^2 \sin \varphi \end{cases}$$

$$\begin{cases} \ddot{\beta} = -0.39 \\ \ddot{c} = -2.3 \end{cases}$$

$$\vec{v}_M = \dot{c} \vec{e}_1 = \dot{c} (\cos \gamma \vec{e}_1 + \sin \gamma \vec{e}_2)$$

$$\vec{a}_M = \ddot{c} \vec{e}_1 = \ddot{c} (\cos \gamma \vec{e}_1 + \sin \gamma \vec{e}_2)$$

$$\frac{dE_c}{dt} = \sum W_i$$

$$E_c = \frac{1}{2} M v_M^2 + \frac{1}{2} m v_G^2 + \frac{1}{2} J \omega^2$$

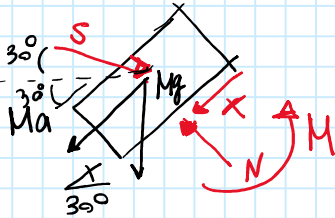
$$\frac{d\vec{E}_C}{dt} = \sum W_i$$



$$E_C = \frac{1}{2} M v_u^2 + \frac{1}{2} m v_a^2 + \frac{1}{2} I \omega^2$$

$$\begin{aligned} \frac{dE_C}{dt} &= M \vec{v}_u \vec{a}_u + m \vec{v}_a \vec{a}_a + I \vec{\omega} \dot{\vec{\omega}} \\ &= M \dot{c} \ddot{c} + m \dot{a} \ddot{a} \frac{a^2}{4} + I \dot{\psi} \ddot{\psi} \end{aligned}$$

$$\sum W_i = C m \dot{\psi} + m \vec{g} \cdot \vec{v}_a + M \vec{g} \cdot \vec{v}_u + \vec{T} \cdot \vec{v}_u$$



$$\begin{cases} \sum \vec{F}_{x'} = 0 \\ \sum \vec{F}_{y'} = 0 \end{cases} \left\{ \begin{array}{l} -T + S \cos 60^\circ - Ma - Mg \sin 30^\circ = 0 \\ N - Mg \cos 30^\circ - S \sin 60^\circ = 0 \\ T = \frac{1}{2} N \end{array} \right.$$

$$\begin{cases} S = 2T + 2Ma + Mg & 131.2 \\ S = \frac{2\sqrt{3}}{3} N - Mg \\ T = \frac{1}{2} N & 39.7 \end{cases} \rightarrow 2 \cdot \frac{1}{2} N + 2Ma + Mg = \frac{2\sqrt{3}}{3} N - Mg$$

$$\rightarrow N \left(\frac{2\sqrt{3}}{3} - 2 \right) = 2Ma + 2Mg$$

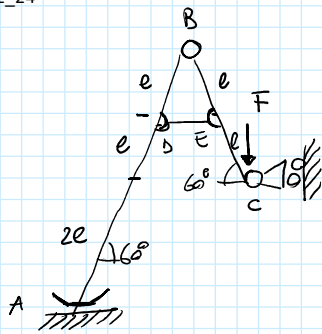
$$N = \frac{2Ma + 2Mg}{\left(\frac{2\sqrt{3}}{3} - 2 \right)} = 198.5$$

$$\sum W_i = C m \dot{\psi} - m g \dot{\psi} \frac{a}{2} \cos \varphi - M \dot{c} \sin \gamma g - T \dot{c}$$

$$C m = \frac{1}{\dot{\psi}} \left[m g \dot{\psi} \frac{a}{2} \cos \varphi + M \dot{c} \sin \gamma g + T \dot{c} + M \dot{c} \ddot{c} + m \dot{a} \ddot{a} \frac{a^2}{4} + I \dot{\psi} \ddot{\psi} \right]$$

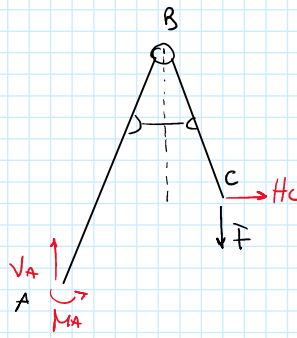
$$C m = -120.45$$

P1es2_24

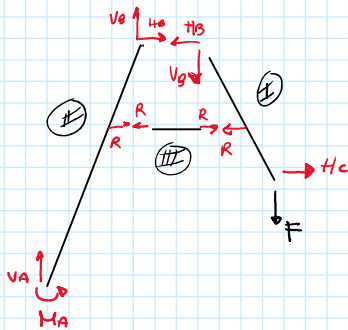


1) Forze in A, B, C, D, E

2) Azioni interne AB

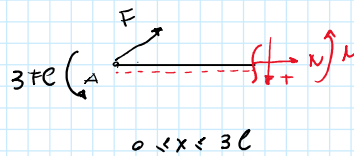
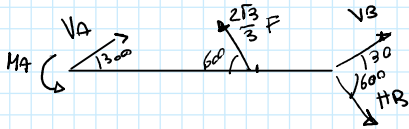


$$\begin{cases} \sum F_x = 0 & \underline{V_A} = F \\ \sum F_y = 0 & \underline{H_C} = 0 \\ \sum M_A = 0 & M_A - F(2l/2 + l/2) = 0 \\ & \underline{M_A} = 3Fl \end{cases}$$

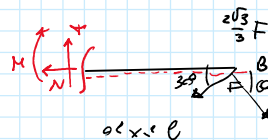


$$\begin{cases} \sum F_y = 0 & \underline{V_B} = -F \\ \sum M_B = 0 & -Fl - Rl \frac{\sqrt{3}}{2} = 0 \\ & \underline{R} = -\frac{2\sqrt{3}}{3}F \\ \sum F_x = 0 & \underline{H_B} = -R = +\frac{2\sqrt{3}}{3}F \end{cases}$$

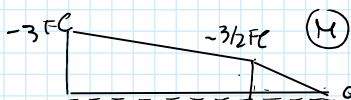
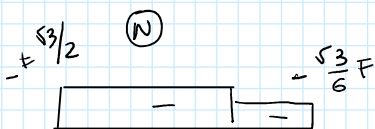
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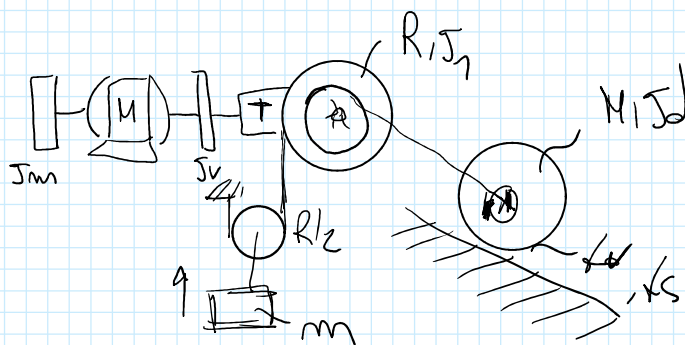


$$\begin{cases} \sum F_x = 0 & N = -F \frac{\sqrt{3}}{2} \\ \sum F_y = 0 & T = F/2 \\ \sum M_A = 0 & H = Tx - 3Fl \end{cases} \begin{cases} M(0) = -3Fl \\ M(3l) = -\frac{3}{2}Fl \end{cases}$$



$$\begin{cases} \sum F_x = 0 & N = +\frac{\sqrt{3}}{3}F - F \frac{\sqrt{3}}{2} = -\frac{\sqrt{3}}{6}F \\ \sum F_y = 0 & T = \frac{F}{2} + \frac{2\sqrt{3}}{3}F \frac{\sqrt{3}}{2} = +\frac{3}{2}F \\ \sum M_B = 0 & H = -Tx = -\frac{3}{2}Fx \end{cases} \begin{cases} M(0) = 0 \\ M(l) = -\frac{3}{2}Fl \end{cases}$$





$$C_{M1} = C_{M0} - A \omega_m$$

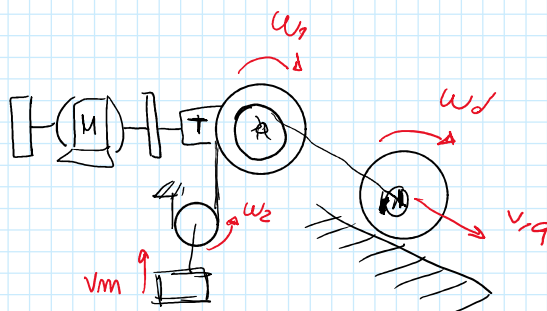
- 1) C_{M1} e ω_m a regime discesa
- 2) C_{M1} discesa per avere $a_d = \times \times \text{ m/s}^2$
- 3) verifica di aderenza del disco al punto ②

$$1) W_m = C_{M1} \omega_m$$

$$W_u = Mg \sin \alpha - mg \sin \alpha - N f_v R \omega_d$$

$$\omega_1 = \gamma \omega_m \quad v_m = \gamma \omega_m \frac{R}{2} \quad \omega_d = \frac{\gamma \omega_m}{2}$$

$$\omega_2 = \gamma \omega_m \quad v = \gamma \omega_m \frac{R}{2}$$



$$W_u = \left(Mg \sin \alpha \gamma \frac{R}{2} - mg \gamma \frac{R}{2} - Mg \cos \alpha f_v \frac{R}{2} \gamma \right) \omega_m$$

$$= (M \sin \alpha - m - M \cos \alpha f_v) g \gamma \frac{R}{2} \omega_m$$

< 0 moto diretto

$$N = Mg \cos \alpha$$

$$W_1 + W_2 + W_p = 0 \quad \eta_D W_m + W_u = 0$$

$$\eta_D C_{M1} \omega_m + (M \sin \alpha - m - M \cos \alpha f_v) g \gamma \frac{R}{2} \omega_m = 0$$

$$C_{M1} = \frac{1}{\eta_D} \left(m + M \cos \alpha f_v - M \sin \alpha \right) g \gamma \frac{R}{2} = 1.72$$

$$C_{M1} = C_{M0} - A \omega_{mR} \rightarrow \omega_{mR} = \frac{C_{M0} - C_{M1}}{A} = 232.7$$

$$② W_1 = C_{M1} \omega_m - (J_m + J_v) \omega_m \dot{\omega}_m$$

$$W_2 = W_u - \left[m \left(\gamma \frac{R}{2} \right)^2 + J_1 \gamma^2 + J_d \frac{\gamma^2}{4} + M \left(\frac{\gamma R}{2} \right)^2 \right] \omega_m \dot{\omega}_m$$

< 0

< 0

$H_p = \omega_m, \dot{\omega}_m > 0$

$$W_2 = W_U - \left[m \left(\frac{vR}{2} \right)^2 + J_1 v^2 + J_0 \frac{v^2}{4} + M \left(\frac{vR}{2} \right)^2 \right] \omega_m \dot{\omega}_m$$

<0

<0

note diretto

$$- J^* \omega_m \dot{\omega}_m$$

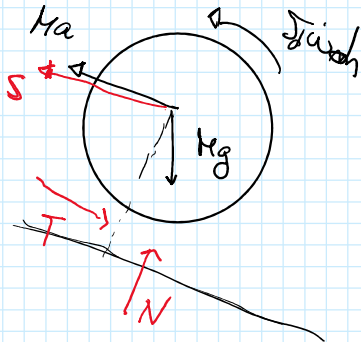
$$\eta_D \text{ con } \omega_m - \eta_D (J_m + J_v) \omega_m \dot{\omega}_m - \left[m \left(\frac{vR}{2} \right)^2 + J_1 v^2 + J_0 \frac{v^2}{4} + M \left(\frac{vR}{2} \right)^2 \right] \omega_m \dot{\omega}_m + W_U = 0$$

$$\eta_D \text{ con } \omega_m + (M \sin \alpha - m - M \cos \alpha f_v) g \frac{v}{2} R \omega_m = J^* \omega_m \dot{\omega}_m$$

$$\text{con} = \frac{1}{\eta_D} \left[(M \cos \alpha f_v + m - M \sin \alpha) g \frac{v}{2} R + J^* \dot{\omega}_m \right] = 21.48$$

$$\dot{\omega}_m = \frac{2a}{vR} = 30$$

③



$$\sum M_c = 0$$

$$J \dot{\omega}_m + TR + N f_v R = 0$$

$$T = - \frac{J \dot{\omega}_m \frac{v}{2} - N f_v R}{R} \quad -3.19$$

$$\sum F_x = 0$$

$$T + Mg \sin \alpha - Ma = S$$

30.85